

CHIDUBEM NDUKWE

✉ chidubemjan31@gmail.com

🌐 linkedin.com/in/ChidubemNdukwe

🌐 duks31.github.io

Summary

Mechatronics engineer who builds autonomous systems end-to-end from embedded control and sensor fusion to perception and navigation. I think in systems, and care about robots that actually work outside the lab and I also build Machine Learning systems that solve problems.

Education

Nile University of Nigeria

B.Eng. Mechatronics Engineering

2021 – 2026

Abuja, Nigeria

Projects

Autonomous Mobile Robot (Cika) | ROS2, RTAB-Map, Nav2, YOLOv8, OAK-D Lite

- Designed and implemented a 4WD skid-steer autonomous mobile robot with a 6-DOF robotic arm for waste detection, collection, and segregation; simulated in Gazebo Ignition Fortress and deployed on a Raspberry Pi 4B.
- Built a YOLOv8n/EfficientNet-B0 perception pipeline on the OAK-D Lite, fused with RPLIDAR C1 and IMU/wheel odometry via EKF, feeding RTAB-Map SLAM and a Nav2 RegulatedPurePursuitController tuned for skid-steer kinematics.
- Ran the full autonomous sequence on physical hardware — detect, navigate, verify, pick — through a state machine driving a Nav2 action client and a UART bridge to ESP32-controlled BTS7960 motor drivers.

Sentinel | Airflow, MLflow, Feast, FastAPI, Docker, PostgreSQL, Redis

- Built a 9-container MLOps fraud detection platform end-to-end, orchestrating Airflow DAGs, a Feast feature store, and MLflow model tracking behind a FastAPI serving layer, achieving 97.7% accuracy with a live prediction API.
- Debugged the full containerised stack from the ground up: inter-container DNS resolution, Airflow DAG scheduling, Feast feature store configuration, and MLflow artifact path handling across PostgreSQL, MinIO, and Redis.

GramAI | Python, llama.cpp, Qwen2.5-3B, RAG

- Built an offline Python traceback debugger using Qwen2.5-3B served locally via llama.cpp, paired with a two-tier RAG system for grounding fixes in known error patterns.
- Achieved 21.56 tokens/sec inference at 3,454 MB peak RAM on local hardware.

Spare Part Demand Forecasting | Python, Scikit-learn, Power BI, DAX, Pandas

- Identified an inventory forecasting gap during an oilfield services internship; built a full data pipeline from raw historical parts data through feature engineering, model training, and validation to a production-ready forecasting system.
- Surfaced predictions through an interactive Power BI dashboard covering \$3.06M in forecasted inventory cost across 828K units, with anomaly flagging for outlier demand patterns.

Car Price Prediction | Scrapy, FastAPI, PostgreSQL, Docker, GitHub Actions, Gradio

- Built an end-to-end data pipeline: Scrapy scraper to PostgreSQL, regression model with hyperparameter tuning, FastAPI serving with Gradio UI.
- Containerised with Docker and automated CI/CD via GitHub Actions for continuous deployment.

Experience

SLB

March 2025 – Sept 2025

Reservoir Performance Intern

Port-Harcourt, Nigeria

- Built a spare part demand forecasting model and Power BI dashboard to address inventory inefficiencies forecasting 828K units and \$3.06M in procurement cost, enabling data-driven stock decisions.
- Performed functional testing of Nitrogen Pump systems, contributing to a 15% reduction in equipment downtime and improved operational safety.
- Collaborated with Mechanical and Electrical Technicians, helping resolve 90% of equipment malfunctions within service deadlines.

Seplat Energy

Aug 2024 – Sept 2024

Information Technology Intern

Abuja, Nigeria

- Re-imaged and configured laptops with IPv4 networking and integrated them into the company domain.
- Deployed and configured new engineering tools for employees, improving team workflow.
- Supported IT infrastructure troubleshooting, resolving employee-reported issues.

Hamoye AI

Aug 2023 – Dec 2023

Data Analytics/Science Intern

Remote

- Developed Python-based data analysis pipelines using Pandas and Scikit-learn to extract insights from complex datasets, supporting data-driven decision-making across project teams.

- Integrated trained data models into existing systems and optimised data processing workflows, improving pipeline reliability and reducing processing latency.

Hamoye AI

Jun 2022 – Oct 2022

Machine Learning Operations (MLOps) Intern

Remote

- Designed and deployed containerised ML workflows using Docker and Kubernetes, establishing reproducible pipelines for model training and inference.
- Automated data pipeline testing and deployment processes, reducing manual intervention and improving integration reliability for analytics workflows in production.

Freelancing

2023 – 2024

Freelance Data Analyst / Scientist

Remote

- Delivered descriptive, inferential, and predictive analyses using Python, SPSS, and Excel for clients across multiple domains.
- Built interactive dashboards in Tableau and Power BI for real-time performance monitoring; automated preprocessing pipelines, reducing manual effort significantly.

Writing & Community

The Nerd Stack

2024 – Present

Technical Writer — Robotics & AI (Substack)

Remote

- Publish in-depth technical articles on robotics, autonomous systems, and AI covering topics such as Vision-Language-Action models, ROS2, and simulation tooling.

Technical Skills

Languages: Python, SQL, C/C++

Robotics & Simulation: ROS2, Gazebo Ignition, RTAB-Map, Nav2, URDF, RViz2, OAK-D Lite, RPLIDAR

ML & CV: PyTorch, TensorFlow, Scikit-Learn, YOLO, OpenCV

MLOps & Data: MLflow, Airflow, Feast, Docker, Docker Compose, Kubernetes, FastAPI, PostgreSQL, Redis, MinIO, GCP

Visualisation & Analytics: Tableau, Power BI, Matplotlib, Pandas, NumPy, Plotly, Streamlit, Databricks, Pyspark

Tools: Git, GitHub Actions, Linux (Ubuntu), VS Code